Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

sustAInX

Team leader name: [Registered team leader name]

Workflow Rescue

Problem Statement: Recover interrupted enterprise workflows while preserving customer intent, permissions and a verifiable outcome.

Brief about the idea

Workflow Rescue is an operator's recovery workspace for an order stranded between services.

A timeout is not the truth.

The charge succeeded. The reservation expired. The customer still expects the original item.

> **“Same item. No extra charge. Two days late is fine.”**

Reconstruct

Read payment, stock and dispatch evidence.

Recover within authority

Gemini proposes; code controls each write.

Prove the result

Independent checks verify or request review.

Synthetic business services. No production customer data.

Opportunities

An evidence-linked recovery layer for the operator who must decide what happens next.

How different is it?

Beyond a retry button: a stock conflict triggers fresh evidence and a revised proposal.

Designed to complement existing workflow engines.

How does it solve it?

Join customer permission, authoritative receipts and guarded execution in one incident.

Uncertainty becomes an explicit handoff.

USP of the solution

Every attempted action has evidence. Every completed recovery has independent checks.

Operators and reviewers inspect the same record.

Target users: order operations • finance reconciliation • engineering reliability • internal control review

Retries, idempotency and compensation already exist. Incremental AI benefit needs broader evaluation.

List of features offered by the solution

One incident, from customer intent to an inspectable business outcome.

01 Your own incident

Edit stock, payment and permissions, or import a validated JSON incident.

02 Evidence gathering

The controller reads payment, inventory and order evidence before planning.

03 Adaptive proposal

Request fresh evidence after a stock-version conflict.

04 Guarded execution

Check typed tools, evidence, versions and idempotency.

05 Independent outcome

Verify state or hand off when payment or dispatch is uncertain.

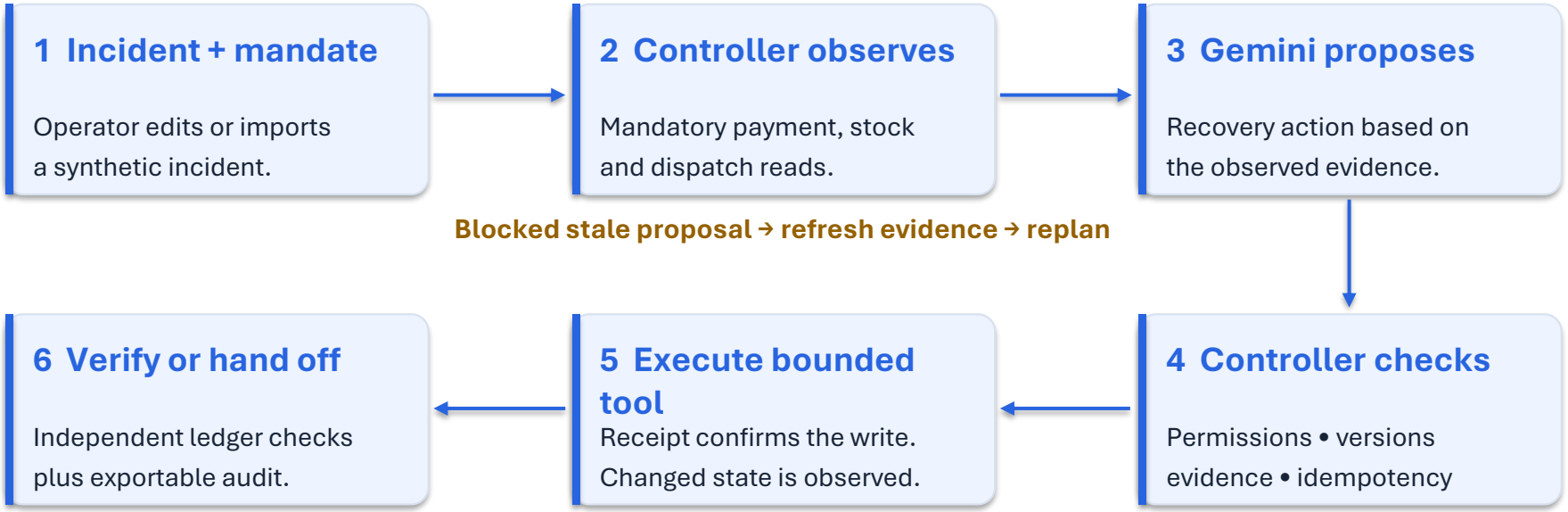
06 Inspectable record

Replay real events, open receipts and export the run's JSON audit.

Replay is read-only playback of recorded events. A fresh run makes new model calls.

Process flow diagram or Use-case diagram

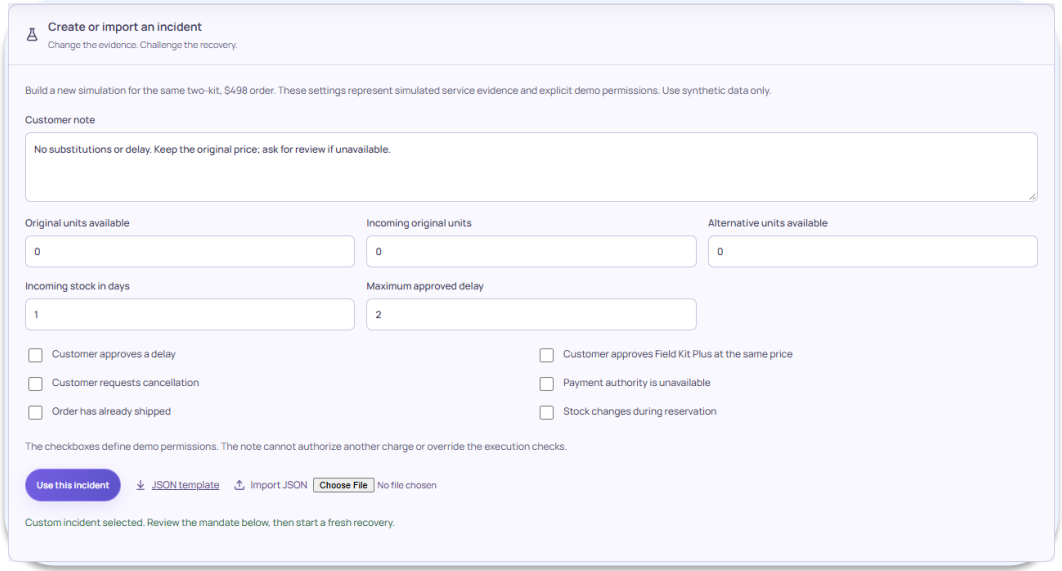
The controller gathers mandatory evidence; Gemini chooses a recovery and adapts to change.



Bounded proposal loop. Unknown payment or irreversible dispatch leads to operator review.

Wireframes/Mock diagrams of the proposed solution (optional)

Implemented incident lab: change evidence, review explicit permissions, then run a new recovery.



Orient

Home introduces the incident and service state.

Act

Edit or import an incident; start a fresh run.

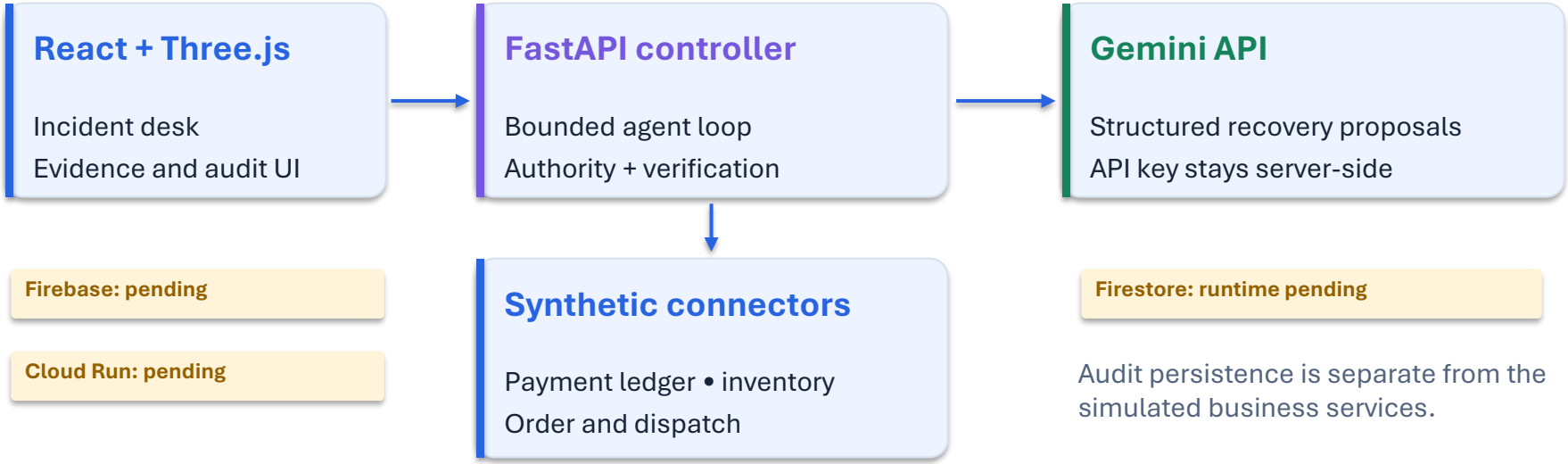
Review

History and receipts reconstruct each run.

Actual application screenshot. Imported JSON is schema-validated; a customer note cannot override permission controls.

Architecture diagram of the proposed solution

Solid boxes: implemented application path. Google deployment status is stated separately.



Current backend: temporary Google Cloud Shell demo. Persistent Cloud Run hosting is not yet deployed.

Technologies to be used in the solution

The Google services have distinct roles; the pitch does not treat planned services as deployed.

Gemini API	Evidence-linked recovery proposal and conflict replanning	Live suite passed
Firebase Hosting	Public submission portal and prototype entry link	Pending deployment
Firestore	Persist completed run evidence for later review	Runtime check pending
Cloud Run	Managed runtime for the FastAPI service	Pending deployment
React / Three.js	Responsive recovery desk, service views and replay	Implemented
FastAPI / Pydantic	Typed proposals, execution guards and verification	Implemented

Verified model: gemini-3.5-flash-lite. No automatic fallback to a paid model.

Only explicit, verified status flags mark a deployment complete. Secrets never enter the frontend or deck.

Estimated implementation cost (optional)

Constrain prototype spend; validate business value with a real operations pilot.

Prototype cost boundary

Build scope: 2–3 day prototype target.

Gemini: user's confirmed Free tier.

Current hosting: free Cloud Shell session; temporary availability.

Persistent hosting and integration costs still need validation.

ILLUSTRATIVE ANNUAL CAPACITY VALUE

$250,000

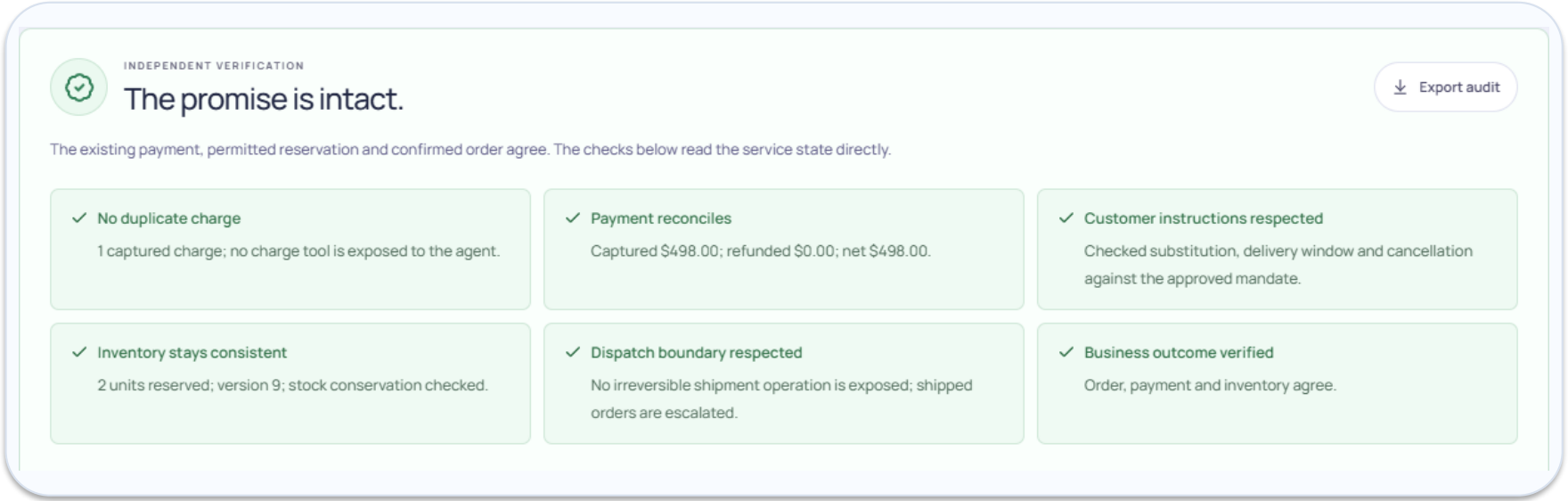
10m orders × 1% recovery incidence
× 50% assisted × 10 minutes saved
× $30/hour ÷ 60

Hypothetical inputs, before cloud, integration and review costs. Released capacity is not automatically cash saved.

No measured ROI, customer adoption or global savings claim. A pilot must validate every assumption.

Snapshots of the prototype

The interface exposes the underlying record; a green animation alone is never proof.



Local deterministic rehearsal; no Gemini calls in this image.

Selected verification checks from an actual run; the caption identifies the provider used.

Prototype Performance report/Benchmarking

Six reviewed development fixtures; useful coverage evidence, not a held-out accuracy benchmark.

6/6 live Gemini **82 Python tests** **8 JavaScript tests**

Scenario	Expected	Gemini outcome	Model calls
stock race	recovered	recovered	4
straight recovery	recovered	recovered	2
cancel order	cancelled	cancelled	3
unknown payment	escalated	escalated	1
approved substitute	recovered	recovered	2
already dispatched	escalated	escalated	1

Rehearsal: 6/6. Baseline escalates on stock conflict; a broader rules engine can solve these fixtures.

gemini-3.5-flash-lite • 0 safety-check failures • 3 controller reads before each model run. Artifact: 2026-10-18T17:13:33.284+00:00

Additional Details/Future Development (if any)

What must happen before real customer orders enter the system.

1 Delivery readiness

Persistent, publicly accessible hosting.

Validate every required Google integration.

Publish source and a working 3–4 minute demo.

2 Customer pilot

Use an owner's sandbox APIs with scoped authority.

Broaden instruction and incident variation.

Measure handling time and safe handoff coverage.

3 Production controls

Service-owned idempotency and durable state.

Tenant identity, access controls and retention.

Tamper-evident records and human approval gates.

Prototype audit is inspectable, but not immutable, certified or independently attested.

Provide links to your:

Public repository, 3–4 minute demonstration, and deployed working product.

GitHub Public Repository — publication pending

https://github.com/abhi14dodiya/workflow-rescue-sustainx

Demo Video Link (3–4 minutes) — upload pending

https://gen-lang-client-0915819772.web.app/downloads/workflow-rescue-demo.mp4

Final Product Link

https://federation-php-architecture-blocked.trycloudflare.com

The current product URL is a temporary session demo, not a durable evaluation deployment.

All links must work without the evaluator signing into the team's private accounts.

Evidence, assumptions and scope

Our claim is controlled, inspectable recovery of synthetic incidents. The pilot must establish business value.

Stripe webhook guidance

Duplicate events establish why recovery must reconcile authoritative state.

Stripe idempotent requests

Safe request retries are an established engineering control.

Temporal sagas and compensation

Partial workflows and application-specific compensation predate this prototype.

Primary technical sources checked 18 October 2026. These sources do not substantiate a product savings figure.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you



Workflow Rescue · sustAInX

Recover the workflow.
Preserve the customer's intent.